NATIONAL INSTITUTE OF TECHNOLOGY PATNA

Department of Computer Science and Engineering

End Semester Examination, July-December 2025

Program & Dept.: B.Tech/B.Tech-M.Tech-DD, CSE
Course Name: Object Oriented Programming
Maximum Time: 3 hours

Semester- III
Course Code: CS34105
Max. Marks: 60

Instructions:

1. *Attempt all questions. Write a concise and precise answer.*
2. *Assume any suitable data, if necessary.*
3. *The marks, Course Outcome (CO), and Bloom's Level (BL) related to questions are mentioned at the end of each question.*

Q.1 What will be the output of the following code snippet/program? *[12M] [CO1,CO2, CO3][BL4, BL5]*

a).
```
int x = 10;
x *= 2 + 3;
System.out.println(x);
```

b).
```
class A {
  int x = 10;
}
class B extends A {
  int x = 20;
}
class Test {
  public static void main(String[] args) {
    A obj = new B();
    System.out.println(obj.x);
  }
}
```

c).
```
String s = "abc";
System.out.println(s.charAt(3));
```

d).
```
byte b = 127;
b++;
System.out.println(b);
```

e).
```
class Test {
  public static void main(String[] args) {
    System.out.println(2 == 2.0);
  }}
```

f).
```
interface P {
  static void m() { System.out.println("P"); }
}
interface Q extends P {
  static void m() { System.out.println("Q"); }
}
public class Test {
  public static void main(String[] args) {
    P.m();
    Q.m();
  }}
```

g).
```
String s = null;
System.out.println(s + "X");
```

h).
```
abstract class A {
  A() { test(); }
  abstract void test();
}
class B extends A {
  int x = 10;
  void test() { System.out.print(x); }
}
public class Test {
  public static void main(String[] args) {
    new B();
  }}
```

i).
```
final int x;
System.out.println(x);
```

j).
```
int x = 7;
System.out.println(x & 3);
```

k).
```
String s = "abcde";
System.out.println(s.substring(2));
```

l).
```
try {
  try {
    int x = 10/0;
  } finally {
    System.out.print("I1");
  }
} catch (Exception e) {
  System.out.print("I2");
}
```

Q.2. **a).** Create three classes A, B, and C in a multilevel inheritance chain. Class A contains two overloaded methods: show(int) and show(Object). Class B overrides show(Object), and class C overrides show(int). Write a program that creates an object of class C and refers to it through A, B, and C references. Explain which version of each method is called and how the binding changes in each case. *[4M][CO2][BL4]*

b). Write a Java program to demonstrate a class that allows creation of only two objects. Ensure that any attempt to create a third object displays the message: *'Can't create the third object'*.

[4M][CO1][BL6]

c). Write a Java program to create a user-defined exception called InvalidMarksException. The program should accept marks of 5 subjects from the user. If any mark is less than 0 or greater than 100, the program should throw InvalidMarksException and display an appropriate error message. Otherwise, it should calculate and display the average of the marks. *[4M][CO3][BL4]*

Q.3. a). Write a Java Program to print the following Pyramid Number Pattern. Number of lines(i.e., 5 in this case) should be entered at run time. *[4M][CO2][BL3]*

```
    1
   2 3 2
  3 4 5 4 3
 4 5 6 7 6 5 4
5 6 7 8 9 8 7 6 5
```

b). Explain the purpose of the 'throws' keyword in Java. Give an example scenario where throws is used. *[2M][CO3][BL2]*

c). Create two interfaces, Playable with a method play() and Recordable with a method record(). Add a default method info() in both interfaces that prints a message about the interface. Implement a class MusicPlayer that implements both interfaces. *[6M][CO2][BL2,BL3,BL4]*

i. Create a MusicPlayer object and call play(), record(), and info().
ii. Show how to access the default info() method of each interface individually from the MusicPlayer class, and explain how Java resolves the conflict.
iii. Explain how the program behavior changes if the info() method in both interfaces is declared as a static method.

Q.4. a). Differenciate between method overriding and method hiding. Write a Java program to demonstrate method hiding. *[4M][CO2][BL4]*

b). Write a Java program to create two threads, where one thread prints "Hello" five times and the other prints "World" five times. The program should demonstrate concurrent execution, so the messages from both threads can appear in any order. *[4M][CO5][BL4]*

c). Write a Java program to read a file and display the frequency of each word in the file. *[4M][CO4][BL3]*

Q.5. a). Write a Java program using Swing to create a simple calculator GUI. The application should accept two numbers from the user using JTextField, provide buttons for addition, subtraction, multiplication, and division, and display the result in a JLabel. Explain how Swing components and event handling make GUI design easier and more flexible compared to AWT. *[6M][CO5][BL3,BL6]*

b). Write a Java program to read a text file using FileInputStream and write its contents to another file using PrintStream. While writing, prefix each line with its line number and convert all text to uppercase. *[6M] [CO4][BL4]*